

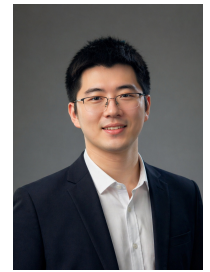


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🌐 <https://ydchen0806.github.io/>

🌐 <https://scholar.google.com/citations?user=hCv1j5cAAAAJ&hl=en&oi=ao>



🎓 EDUCATION

University of Science and Technology of China & Shanghai AI Lab 📍 Hefei & Shanghai

Information and Communication Engineering, Ph.D. Candidate

2024.7 ~ 2027.6 (Expected)

- **Research Focus:** Predictive multimodal intelligence: from representation and predictive learning to world models for embodied perception, future prediction, and action generalization
- **Advisors:** Prof. Feng Wu (CAE Academician, IEEE Fellow), Prof. Zhiwei Xiong; Co-supervisor: Prof. Xiaoou Tang (IEEE Fellow, Shanghai AI Lab)
- **Core Courses:** Algorithm Design and Analysis, Statistical Learning, Deep Learning, Reinforcement Learning
- **Honors:** Principal Investigator of National Natural Science Foundation Ph.D. Project (2024)

University of Science and Technology of China 📍 Hefei

Computer Technology, M.S.

2022.9 ~ 2024.7

- **Honors:** National Graduate Scholarship (2022)

Xiamen University 📍 Xiamen

Environmental Ecological Engineering & Economics (Dual Degree), B.S.

2018.9 ~ 2022.7

- **Overall Ranking:** 1/31
- **Honors:** Xiamen University Academic Star (2021), CDA Level 1 Certification (2022), Kaggle Expert
- **Advisor:** Prof. Yuanze Zhang

🔬 RESEARCH PROJECTS

Embodied Intelligence & World Models: Pelican Series

2026.5 ~ Present

Unified Embodied Foundation Models *Core Contributor (Ranked Second); VLA Technical Report*

- In **Pelican-Unified 1.0**, helped build a unified embodied model for **understanding, reasoning, imagination, and action**, integrating multimodal understanding, future video rollout, and low-level action generation.
- As the concrete **VLA component** within the Pelican-Unified framework, contributed to the **Pelican-VLA 0.5: Attending Before Acting Benefits Generalization** technical report, unifying vision-language understanding, **future-frame generation**, and action prediction; learnable **Bottleneck Tokens** support cross-scene and cross-embodiment generalization.
- Together, these technical reports connect **unified multimodal representations, future-state modeling, and closed-loop action** into a world-model narrative. Pelican-Unified 1.0; Pelican-VLA 0.5
- Media coverage: People's Daily, Beijing Daily / Beijing Municipal Government, Machine Heart, Zhidongxi

National Natural Science Foundation Ph.D. Project

2025.1 ~ 2027.12

NSF Project *Principal Investigator (Funding: 300k RMB, Only recipient in Information field, Anhui Province, 2024)*

- [1] Foundation **10-billion-scale** brain neural **foundation model** training
 - Neuron underlying visual feature **token construction**
 - Neural architecture search, **autoregressive visual model** training
- [2] **Fine-grained** neuron segmentation
 - **Skeleton features**, attention decomposition mechanisms
 - Complementary masking, segmentation **robustness enhancement**
- [3] Neural **circuit reconstruction**
 - Point cloud morphological features and image feature **alignment**
 - Candidate competition strategy, **progressive reasoning**, long-range tracking

Representation and Predictive Learning

2022.5 ~ 2023.12

Conference&Journal *First Author*

- [1] **TokenUnify: Scalable Autoregressive Visual Pre-training with Mixture Token Prediction, ICCV 25** **[CCF-A]**
 - Proposed image autoregressive pretraining combined with Mamba framework, demonstrating advantages of long sequences and low computational cost.
 - Demonstrated good scaling laws with corresponding theoretical proof, showing logarithmic-linear relationship between model performance and parameters.
- [2] **MaskTwins: Dual-form Complementary Masking for Domain-Adaptive Image Segmentation, ICML 25** **[CCF-A]**
 - Proposed new complementary masking theory from sparse signal reconstruction perspective, rigorously proving theoretical advantages of dual-form complementary masking in extracting domain-invariant features.
 - Designed simple and efficient unsupervised domain adaptation framework, achieving cross-domain image segmentation through complementary masking consistency learning, improving natural image segmentation by 2.7% mIoU and biomedical image segmentation by 3.2% IoU.

- [3] **[LONG ORAL] Self-supervised neuron segmentation with multi-agent reinforcement learning, IJCAI 23 [CCF-A]**
 - Improved MAE masking strategy using reinforcement learning, automatically selecting mask ratios and schemes.
 - Introduced multi-agent framework for more efficient self-supervised feature learning, improving segmentation accuracy by 12%.
- [4] **Learning multiscale consistency for self-supervised electron microscopy instance segmentation, ICASSP 24 [CCF-B]**
 - Achieved high-performance pretraining strategy based on multiscale feature contrastive learning and feature reconstruction.
 - Proposed feature consistency loss function, overcoming representation differences caused by scale variations, improving accuracy by 9%.

Multimodal and Generative Models

2023.8 ~ Present

Conference&Journal First Author & Co-first Author

- [1] **[Accepted] Learned Image Coding with Generative Reference of Conditional Latents, TPAMI [SCI-Q1]**
 - Extension work of AAAI 25 oral paper, further exploring benefits of reference images for image coding.
 - Obtained reference images through local dictionary, network retrieval, and image generation, improving compression performance by 15%.
 - Proposed conditional latent generation framework, effectively solving performance degradation when reference images are unavailable.
- [2] **[ORAL] Conditional Latent Coding with Learnable Synthesized Reference for Deep Image Compression, AAAI 25 [CCF-A]**
 - Built image similarity dictionary for retrieving similar images to improve entropy model probability estimation.
 - Proposed learnable synthetic reference framework, improving PSNR by 0.6dB at the same bit rate with leading compression performance.
- [3] **[Major Revision] Generative Text-Guided 3D Vision-Language Pretraining for Unified Medical Image Segmentation, Submit to PR**
 - Generated image descriptions using large language models for multimodal image-text contrastive learning pretraining.
 - Introduced generative text guidance to address scarce medical-image annotations and enable zero-shot segmentation.
- [4] **BIMCV-R: A Landmark Dataset for 3D CT Text-Image Retrieval, MICCAI 24 [CCF-B]**
 - Built the first open-source 3D CT image-text pair dataset containing 10,000 high-quality medical image-description pairs.
 - Achieved efficient image-text information retrieval and keyword search, improving recall rate by 25% over traditional methods.
- [5] **MaskFactory: Towards High-quality Synthetic Data Generation For Dichotomous Image Segmentation, NeurIPS 24 [CCF-A]**
 - Generated corresponding mask-image pairs using ControlNet through rigid and non-rigid deformation editing of masks.
 - Core techniques: deformation-driven mask editing, ControlNet-guided image generation, synthetic data quality assessment.
 - Synthetic data performance approached real data in downstream segmentation tasks with only 2% performance gap, significantly reducing annotation costs.
- [6] **[Major Revision] Joint Semantic and Coded Generation for Conditional Latent Coding, Submit to TCSVT**
 - Proposed joint semantic and coded generation framework, combining text descriptions with minimal coded features (0.01 bpp) to guide controllable rectified flow models in generating high-fidelity reference images.
 - Core techniques: ControlNet for DiT architecture, iterative reference alignment (IRA), dynamic feature fusion (DFS), iterative probability refinement (IPR), theoretical robustness proof.
 - Improved compression performance by 15.8% BD-rate compared to VVC, successfully unifying image generation and compression, bridging the gap between semantic and coded representations.
- [7] **EMPOWER: Evolutionary Medical Prompt Optimization With Reinforcement Learning, JBHI [SCI-Q1]**
 - Proposed the first evolutionary prompt optimization framework for medical domain, combining domain knowledge and reinforcement learning.
 - Core techniques: medical attention, multi-dimensional quality assessment, evolutionary search, and semantic verification for clinical guideline consistency.
 - Reduced factual errors by 24.7%, improved domain specificity by 19.6%, and achieved 15.3% higher clinician preference in blind testing.

Large Model Engineering

2023.9 ~ Present

Projects Core Member

- **Foundation-model systems:** led a billion-parameter image-coding architecture with 30% higher compression than traditional standards; built LLM-driven quantitative-research workflows with 10k-GPU-scale training experience.
- **Embodied and biomedical AI:** contributed to real-robot closed loops, Unify pretraining, and embodied-AI infrastructure on 100-GPU H200 clusters; led medical-image and neuron-model pretraining with 64 A40 GPUs, DDP, and DeepSpeed for 30B-parameter-scale optimization.

INDUSTRY INTERNSHIP EXPERIENCE

Ubiquant

📍 Beijing

2026.5 ~ Present

Quantitative Research and Large Language Models Researcher

- Working on **quantitative research** and **general large models**, targeting **DeepSeek**-level capabilities.

- Experienced with **10k-GPU-scale clusters** and exploring LLM applications in factor mining and data-driven decision making.

Beijing Humanoid Robot Innovation Center (UBTECH) 📍 Beijing 2025.12 ~ Present

Embodied Intelligence World Model Algorithm Team *Core Member*

- Developed **unified embodied world models** for humanoid robots, connecting task-relevant perception, future-state prediction, and low-level action generation in real-robot closed loops.
- Contributed to **Unify** pretraining and planning validation; investigated its VLA bottleneck representations for cross-scene and cross-embodiment manipulation generalization.
- Experienced with **100-GPU H200** cluster training for large-scale embodied foundation models.

Tencent (IEG) 📍 Shanghai 2025.8 ~ 2025.12

Qingyun Program *Research Intern*

- Worked on **game world models** for game scene understanding, behavior modeling, and content generation.
- Delivered the **Honor of Kings Lingbao** project, focusing on **TTS** and **real-time match commentary**.
- Trained and optimized Lingbao speech generation models based on f5-tts and indextts2.

PLA General Hospital (301 Hospital) 📍 Beijing 2023.9 ~ 2024.2

Data Compression Group *Research Intern*

- Collaborated with Academician Qionghai Dai's team on efficient data compression research.
- Designed medical imaging-specific compression algorithms optimized for CT, MRI modalities, improving compression efficiency by 35%.

🏛️ ACADEMIC INTERNSHIP EXPERIENCE

Imperial College London 📍 London (Remote) 2022.11 ~ 2023.8

Data Science Institute *RA*

- Collaborated with Associate Professor Rossella Arcucci on multimodal pretraining research, submitting one journal paper.
- Developed image-text contrastive learning framework, achieving 93.5% accuracy on medical diagnosis tasks.

Xiamen University WISER Club 📍 Xiamen 2021.8 ~ 2022.7

Data Mining Group *Academic Community Member*

- Responsible for designing and discussing data mining courses, teaching clustering and Transformer sections.
- Mentored 20 undergraduate students to complete machine learning projects, organized 2 campus competition activities.

Wang Yanan Institute of Economics, Xiamen University 📍 Xiamen 2020.8 ~ 2021.12

Econometrics *Research Assistant*

- Assisted Associate Professor Jiong Zhu in completing territorial economic statistics, conducting visual feature extraction of homestead information.
- Developed satellite image analysis tools for automatic land use change identification with 85% accuracy.

🏆 HONORS AND AWARDS

- **National Natural Science Foundation Ph.D. Project** 2024.12
– Principal Investigator, Only recipient in Information field, Anhui Province
- **Interdisciplinary Contest in Modeling (ICM), Outstanding Winner** 2024.05
– Top 0.17% of 10,388 participating teams worldwide
- **National Graduate Scholarship** 2022.12
– Award rate 1%
- **Xiamen University Academic Star** 2021.12
– Only undergraduate recipient
- **'Jingrun Cup' Mathematics Competition** 2021.09
– First Place, Xiamen University
- **'Internet+' Innovation Competition** 2021.08
– Gold Award, Fujian Province
- **National College Student Mathematics Competition Finals** 2021.05
– National Second Prize
- **'Challenge Cup' Academic Competition** 2021.05
– First Prize, Fujian Province
- **National College Student Mathematics Competition** 2020.11
– First Place, Fujian Province

⚙️ TECHNICAL SKILLS

- **Programming:** Python, C, C++, Java, MATLAB, $\LaTeX$
- **Deep Learning:** PyTorch, TensorFlow, DeepSpeed, DDP
- **Languages:** English (TOEFL 110, GRE 328), Chinese (Native)
- **Tools:** Git, Docker, CUDA, HPC

Service: Reviewer for CVPR'25, ICCV'25, WACV'26, NeurIPS'24, ICML'25, ICLR'24, MICCAI'25, ACM MM'24, AISTATS'24, IJCV, and TIP.